

Date	24 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Identify and document the requirements of the OptiCrop system to clearly specify its functionalities and performance expectations. Functional Requirements describe the features and services that the system must provide, while Non-Functional Requirements define the quality attributes, constraints, and performance standards that ensure the system operates efficiently, reliably, and securely.

Functional Requirements (FR): Define the core features and operations of the OptiCrop application, such as crop recommendation, input validation, prediction generation, and result visualization.

Non-Functional Requirements (NFR): Define the performance, scalability, security, reliability, usability, and maintainability requirements that the OptiCrop system must satisfy to deliver a high-quality user experience.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the application without registration. Future versions may support secure login for personalized services.	Medium

2	Authorization levels	Different access levels can be provided for Farmers, Administrators, and Agricultural Experts to manage recommendations and system data.	Medium
3	External interfaces	The system interacts with trained Machine Learning models, datasets, and future integrations such as weather APIs and agricultural databases.	High
4	Transactions processing	The system receives soil and environmental parameters, validates inputs, processes them through the ML model, and generates crop recommendations in real time.	High
5	Reporting	The system displays crop recommendations, confidence scores, alternative crop suggestions, and analytical visualizations for users.	High
6	Business rules, etc.	Crop recommendations must be generated only from valid soil and environmental inputs. The system follows predefined model logic and prediction rules.	High

7	Compliance to laws or regulations	The system must ensure responsible handling of user data and comply with applicable data privacy and ethical AI guidelines.	Medium
8	<i>Other</i>	The application should provide a responsive user interface, AI assistant support, and future scalability for additional agricultural services.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should generate crop recommendations quickly after receiving user inputs.	Prediction results displayed in less than 2 seconds .
2	Scalability	The system should support increasing numbers of users and future feature enhancements without significant performance degradation.	Supports 500+ concurrent users with consistent response times.
3	Security & Data Privacy	User inputs and application data should be protected from unauthorized access and misuse.	Secure data handling with input validation and HTTPS support .

4	Reliability & Availability	The application should remain operational and accessible with minimal downtime.	99% uptime and stable prediction service availability.
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use for farmers and agricultural stakeholders.	Responsive design supporting desktop, tablet, and mobile devices .
6	Other	The system should be easy to maintain, update, and deploy across different environments.	Compatible with Windows, Linux, and cloud deployment platforms such as Render.